Machine Learning Engineer Assessment Report

Validation and Data Split Approach

To validate the model, the 'data/train_test.csv' dataset was split into a training set and a testing set using an 80/20 split ratio. The split was performed using a random seed for reproducibility. Since the dataset is large enough, an 80/20 split provides a sufficient amount of data for the LightGBM model to learn patterns while holding out a representative sample for evaluation.

The target variable is 'posted_rate'. Key features were extracted such as 'month', 'day_of_week', and 'day_of_year' from the 'date' column. Categorical features ('pickup', 'delivery', 'equipment') were passed directly into LightGBM to utilize its built-in categorical handling capabilities. Features missing from the December chart inputs (like lat/lon and market signals) were removed to ensure the model could generalize to the December predictions.

December Prediction Chart

